

MODULE *MCtrace_fix*

SCRIPTED reachability witness: drive the REAL ccraft actions, in a fixed order, from the real *Init* to a state that violates *LeaderCompletenessInv*. A *step counter* enables exactly one action per step, so every state is reached by genuine ccraft transitions(messages are produced by the spec, not fabricated).

Story(3 nodes, quorum = 2, StartTerm = 2; start leader assumed n1) :

- n1@2 appends a client entry e2 and signs it(s2), but does NOT replicate it.
- n2 times out, wins term 3(n3 still only has the start log, so the up-to-date check lets n2 win), then appends/sign its own e3/s3 → n2 DIVERGES at index 5.
- n1(steppeped down to Follower@3 by n2 s vote request) times out, wins term 4 with n3; BecomeLeader truncates n1 to its committable prefix, which KEEPS e2/s2.
- n1 appends e4/s4, replicates idx 5..8 to n3 (one NACK backup, then fills).
- AdvanceCommitIndex(n1) commits the term-4 signature (idx 8), INDIRECTLY committing the term-2 entry e2 (idx 5). n2 is still Leader@3 with a term-3 entry at idx 5.
- ⇒ LeaderCompletenessInv breaks.

EXTENDS *ccfix*, *TLC*

CONSTANTS *NodeOne*, *NodeTwo*, *NodeThree*

ToServers $\triangleq$ {*NodeOne*, *NodeTwo*, *NodeThree*}

Configurations $\triangleq$ ⟨*ToServers*⟩

VARIABLE *step*

varsS $\triangleq$ ⟨*vars*, *committed*, *step*⟩

TInit $\triangleq$

∧ *step* = 0

∧ *committed* = {}

∧ *InitMessagesVars*

∧ *InitCandidateVars*

∧ *InitLeaderVars*

∧ *preVoteStatus* = [*i* ∈ *Servers* ↦ {*PreVoteDisabled*}]

∧ *InitLogConfigServerVars*(*Configurations*[1], *JoinedLog*)

multi-node start (*JoinedLog*)

Each entry: < *stepGuard*, Action > . Action is a REAL ccraft action with pinned params.

Move(*k*, *A*) $\triangleq$ *step* = *k* ∧ *A* ∧ *committed*' = *committed* ∪ *LeaderCommitCerts* ∧ *step*' = *k* + 1

TNext $\triangleq$

∨ *Move*(0, *ClientRequest*(*NodeOne*))

e2 @ idx5 (term2)

∨ *Move*(1, *SignCommittableMessages*(*NodeOne*))

s2 @ idx6 (term2)

∨ *Move*(2, *Timeout*(*NodeTwo*))

n2 → Candidate@3

∨ *Move*(3, *RequestVote*(*NodeTwo*, *NodeThree*))

∨ *Move*(4, *RequestVote*(*NodeTwo*, *NodeOne*))

∨ *Move*(5, *Receive*(*NodeThree*, *NodeTwo*))

n3 UpdateTerm → 3

∨ *Move*(6, *Receive*(*NodeThree*, *NodeTwo*))

n3 grants vote

∨ *Move*(7, *Receive*(*NodeOne*, *NodeTwo*))

n1 UpdateTerm → 3 (steps down)

$\vee \text{Move}(8, \text{Receive}(\text{NodeOne}, \text{NodeTwo}))$
 $\vee \text{Move}(9, \text{Receive}(\text{NodeTwo}, \text{NodeThree}))$
 $\vee \text{Move}(10, \text{Receive}(\text{NodeTwo}, \text{NodeOne}))$
 $\vee \text{Move}(11, \text{BecomeLeader}(\text{NodeTwo}))$
 $\vee \text{Move}(12, \text{ClientRequest}(\text{NodeTwo}))$
 $\vee \text{Move}(13, \text{SignCommittableMessages}(\text{NodeTwo}))$
 $\vee \text{Move}(14, \text{Timeout}(\text{NodeOne}))$
 $\vee \text{Move}(15, \text{RequestVote}(\text{NodeOne}, \text{NodeThree}))$
 $\vee \text{Move}(16, \text{Receive}(\text{NodeThree}, \text{NodeOne}))$
 $\vee \text{Move}(17, \text{Receive}(\text{NodeThree}, \text{NodeOne}))$
 $\vee \text{Move}(18, \text{Receive}(\text{NodeOne}, \text{NodeThree}))$
 $\vee \text{Move}(19, \text{BecomeLeader}(\text{NodeOne}))$
 $\vee \text{Move}(20, \text{ClientRequest}(\text{NodeOne}))$
 $\vee \text{Move}(21, \text{SignCommittableMessages}(\text{NodeOne}))$
 $\vee \text{Move}(22, \text{AppendEntries}(\text{NodeOne}, \text{NodeThree}))$
 $\vee \text{Move}(23, \text{Receive}(\text{NodeThree}, \text{NodeOne}))$
 $\vee \text{Move}(24, \text{Receive}(\text{NodeOne}, \text{NodeThree}))$
 $\vee \text{Move}(25, \text{AppendEntries}(\text{NodeOne}, \text{NodeThree}))$
 $\vee \text{Move}(26, \text{Receive}(\text{NodeThree}, \text{NodeOne}))$
 $\vee \text{Move}(27, \text{Receive}(\text{NodeOne}, \text{NodeThree}))$
 $\vee \text{Move}(28, \text{AppendEntries}(\text{NodeOne}, \text{NodeThree}))$
 $\vee \text{Move}(29, \text{Receive}(\text{NodeThree}, \text{NodeOne}))$
 $\vee \text{Move}(30, \text{Receive}(\text{NodeOne}, \text{NodeThree}))$
 $\vee \text{Move}(31, \text{AppendEntries}(\text{NodeOne}, \text{NodeThree}))$
 $\vee \text{Move}(32, \text{Receive}(\text{NodeThree}, \text{NodeOne}))$
 $\vee \text{Move}(33, \text{Receive}(\text{NodeOne}, \text{NodeThree}))$
 $\vee \text{Move}(34, \text{AppendEntries}(\text{NodeOne}, \text{NodeThree}))$
 $\vee \text{Move}(35, \text{Receive}(\text{NodeThree}, \text{NodeOne}))$
 $\vee \text{Move}(36, \text{Receive}(\text{NodeOne}, \text{NodeThree}))$
 $\vee \text{Move}(37, \text{AdvanceCommitIndex}(\text{NodeOne}))$

$n1$ rejects (more up-to-date)
 $n2$ tallies $n3$'s grant
 $n2$ drains $n1$'s reject
 $n2$ Leader@3 (truncates to $idx4$)
 $e3 @ idx5$ ($term3$) – DIVERGENCE
 $s3 @ idx6$ ($term3$)
 $n1 \rightarrow \text{Candidate}@4$

 $n3 \text{ UpdateTerm} \rightarrow 4$
 $n3$ grants $n1$
 $n1$ tallies grant
 $n1$ Leader@4 (truncate KEEPS $e2/s2$)
 $e4 @ idx7$ ($term4$)
 $s4 @ idx8$ ($term4$)
 $AE idx7 \rightarrow n3$ rejects (gap)
 $n3$ NACK ($lastLogIndex = 4$)
 $n1$ backs $sentIndex$ to 4
 $AE idx5$ ($e2$)
 $n3$ appends $e2$
 $matchIndex = 5$
 $AE idx6$ ($s2$)
 $n3$ appends $s2$
 $matchIndex = 6$
 $AE idx7$ ($e4$)
 $n3$ appends $e4$
 $matchIndex = 7$
 $AE idx8$ ($s4$)
 $n3$ appends $s4$
 $matchIndex = 8$
 $commit idx8 \rightarrow$ indirectly $idx5 \Rightarrow VIOLATION$

$$\text{TraceSpec} \triangleq TInit \wedge \square[TNext]_{varsS}$$